

HW 2: AE 319

Problem 1

Consider a two-dimensional, incompressible flow over an airfoil in the x-y plane. The velocity components in this flow are given by:

$$u = \frac{Cy}{(x^2 - y^2)}$$
$$v = \frac{Cx}{(x^2 - y^2)}$$

where C is a constant.

For this velocity, perform the following:

1. Prove that this flow is physically possible.
2. Derive the equation for the streamlines of the flow.
3. Using a computational tool such as MATLAB or Python, plot the streamlines to visualize the flow pattern. Also, indicate arrows showing the flow direction in each quadrant.
4. If $C = 0.25$, calculate the velocity vector and magnitude for the fluid at point $(x, y) = (4, 1)$.

Problem 2

(Note: some parts of this problem are a worked-out example in your textbook. Use that as a guide — do not blindly copy.) Consider the incompressible, irrotational, two-dimensional flow where the stream function is given by:

$$\psi = 4xy$$

- (a) Determine the velocity field.
- (b) Prove that this flow is both physically possible and irrotational.
- (c) Calculate and graph (do not sketch, use a plotting utility such as MATLAB) the streamline pattern.

- (d) Find the velocity potential for this flow and graph the lines of constant potential on the same graph as the streamlines. Make sure the lines are clearly distinguished using line style or color.
- (e) (Bonus) Use the `quiver` function in MATLAB and plot the velocity vectors.

Problem 3

Please go through part of the section on “Flow lines and flow visualization” (pages 59-78) in the online book *Multimedia Fluid Mechanics* (link to access the book, using your ERAU login is given on Canvas). If you are off campus, you will have to go to the Hunt Library webpage, search for the book in the search utility, and then log in to access the book. After going through this section, please answer the following questions:

- (a) Give six specific flow visualization techniques.
- (b) What is a timeline? Explain with a sketch.